

Leveraging momentum clustering and PID control for enhanced portfolio management

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ABSTRACT

Efficient wealth management in dynamic financial environments is crucial for maximizing returns and mitigating risk. Therefore, this study presents an adaptive portfolio management strategy that combines momentum-based clustering, a Proportional-Integration-Derivative (PID) controller, and efficient frontier (EF) analysis to construct optimal portfolios in dynamic financial environments. A novel integration of a PID controller dynamically adjusts the asset weights to address the limitations of static optimization methods. EF analysis is then employed to identify portfolios with optimal risk-return trade-offs. Our dynamic strategy demonstrates competitive performance compared to conventional benchmarks. From 2017 to 2023, the approach yielded a cumulative return of 135.5%, surpassing Standard and Poor's 500 (46.66%), Russell's 2000 Index (11.47%) and NASDAQ (67.88%). These findings suggest that the integration of clustering techniques and PID controllers offers a promising approach for portfolio management.

IMPACT STATEMENT

This research presents a novel framework integrating momentum clustering and PID control into portfolio management, leveraging a decade of S&P 500 stock data. By combining technical indicators, rolling factor betas, and dynamic optimization, the strategy demonstrates superior adaptability to varying market conditions. The incorporation of advanced metrics, such as Sharpe and Sortino ratios, and efficient frontier modeling ensures rigorous evaluation of performance. This approach outperforms traditional benchmarks like the S&P 500 and DJIA and provides a scalable and robust solution for achieving optimal risk-adjusted returns. The findings contribute significantly to the fields of quantitative finance and algorithmic trading by offering a methodology that bridges static optimization with real-time adaptability, addressing the challenges of economic cyclicity and market volatility.

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1. Introduction

Efficient wealth management in dynamic financial environments represents a fundamental challenge for investors seeking to achieve optimal returns while managing risk. Given the complexities inherent in modern markets, traditional investment strategies often fall short of adapting to rapidly changing conditions. Therefore, this study proposes an innovative approach that integrates momentum clustering, Proportional-Integral-Derivative (PID) controller and Efficient Frontier (EF) analysis to enhance portfolio performance. The significance of this study lies in its aim to address the dual objectives of maximizing returns and minimizing risks in dynamic financial markets. By leveraging a combination of advanced analytical techniques and control mechanisms, this approach offers a systematic framework for navigating market uncertainties and capitalizing on emerging opportunities.

Clustering techniques and algorithms are widely used in financial management. These techniques help sort assets into categories and select the best assets within each category (Theron & van Vuuren, 2018). Hierarchical clustering algorithms have been shown to generate portfolios with reduced volatilities and

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increased stability compared to traditional Mean-Variance Optimization (MVO) (Lu et al., 2018). The proposed method utilizes data mining techniques, specifically the k-means algorithm, to analyze anomalies within accounting statements. This approach parallels the effectiveness observed in portfolio investments, where hierarchical clustering methods have demonstrated utility. In addition, portfolios constructed with unweighted modular components often outperform those that rely on weighted modular components (Le-Khac, 2012). Clustering algorithms offer versatile applications ranging from classifying and analyzing strategic risks to scrutinizing anomalies within accounting statements and pinpointing abnormal financial behavior in companies. For instance, they can be leveraged to delve into diversification development and provide insightful case studies for risk analysis (Wen, 2022). This review covers various clustering methods in finance, including K-means, Minimum Spanning Tree (MST) and hierarchical approaches. These methods, ranging from K-means to MST and hierarchical clustering, find utility in credit scoring, stock market analysis, portfolio selection and trading strategy development (G. Tang et al., 2022).

Clustering methods are frequently employed in finance for multiple purposes including fraud detection, credit evaluation and understanding household heterogeneity. These techniques help identify patterns and clusters in financial data, which can provide valuable insights for decision-making. Clustering methods are used to group companies listed on the Tehran Stock Exchange and forecast changes in stock prices. By employing techniques such as K-means, MST and hierarchical clustering, companies are organized based on shared financial attributes (Li et al., 2022). Deep learning-based clustering is used to investigate high-dimensional household heterogeneity based on balance sheet data (Nourahmadi et al., 2022). Clustering algorithms for investor segmentation in the financial sector were compared (Hwang et al., 2023). These methodologies determine abnormal financial behavior and analyze anomalies in accounting statements using data mining methods, which is a potential area for future research (Bay et al., 2022). Introduce novel methodologies and approaches such as hybrid clustering algorithms, modified versions of traditional algorithms and integrated approaches to address the challenges in financial clustering.

The novel financial clustering methodology incorporates various advanced approaches, including hybrid clustering algorithms, modified versions of traditional algorithms, integrated methodologies and advanced feature selection techniques. These diverse approaches are geared toward improving the accuracy, robustness, interpretability and efficiency of financial clustering methods (D'Urso et al., 2021). Hybrid clustering algorithms leverage diverse techniques to boost accuracy and robustness, whereas integrating clustering with optimization methods enhances the interpretability of financial clusters. Advanced feature selection methods have been employed to enhance the efficiency and effectiveness of financial clustering algorithms. Together, these integrated approaches aim to increase the accuracy, robustness, interpretability and efficiency of financial clustering methodologies (Terraza & Toque, 2021). The correlation block model clustering approach in financial asset selection proves effective by pinpointing a small set of stocks that mirrors the diversification level of the entire stock universe (W. Tang et al., 2022). Overall, clustering techniques play a crucial role in various aspects of finance management strategies, offering insights and aiding decision-making processes.

A momentum refers to the phenomenon in which the past price performance of a security is related to its future returns. Momentum strategies generate higher returns for investors (Huang et al., 2023). Optimized momentum trading strategies encompass various techniques, including utilizing Moving Average Convergence-Divergence (MACD) oscillators alongside n -day holding strategies (Zhang et al., 2022). Additionally, the optimal portfolio weight in momentum strategies depends not only on momentum, but also on the historical price path, with different paths leading to different optimal weights (Li & Liu, n.d.). These strategies have demonstrated varying levels of success across different asset classes, including currencies and cryptocurrencies (Rohrbach et al., n.d.). Overall, modern momentum trading strategies offer investors the potential to diversify portfolios, hedge risks, and generate positive alpha.

Modern portfolio management strategies have evolved over the years. A classic approach rooted in the Efficient Market Hypothesis serves as a foundational framework for understanding market behavior (Akhiezer et al., 2023). Modern Portfolio Theory (MPT) emphasizes the prioritization of low risk and diversification over high returns (Wen, 2023). The effectiveness of the MPT has been evaluated and is generally found to be beneficial (Yu & Zhang, 2023). Portfolio optimization, utilizing predictions from Deep Neural Networks (DNNs), aims to augment investor profitability by merging advanced technology with MPT. Three types of DNNs are employed: Convolutional Neural Network (CNN), Long Short-Term

Memory (LSTM) neural network and Deep Multilayer Perceptron, which seek to refine portfolio decision-making through sophisticated computational methods (Ma, Han, & Wang, 2020).

Machine learning (ML) has revolutionized portfolio management by optimizing asset allocation, improving risk management and enhancing decision-making. ML techniques, such as reinforcement learning (RL) and deep learning, enable dynamic, data-driven investment strategies that adapt to market fluctuations. These systems process vast financial datasets to adjust allocations in real time, boosting returns while managing risks. For example, an AI-driven system achieved an annual Return on Investment of 12.5% and a Sharpe Ratio (SR) of 1.2 (Ambuli et al., 2024). Methods like Deep Deterministic Policy Gradient and Proximal Policy Optimization continuously learn and adapt, outperforming traditional approaches (Daniel et al., 2024). Recent models improve algorithmic transparency, addressing the 'black-box' issue and providing actionable insights for traders (Kottooru et al., 2024). Combining rule-based strategies with ML enhances explainability and risk-adjusted performance (Choi & Kim, 2024). Despite its potential, ML in portfolio management still faces challenges, particularly in ensuring model interpretability and maintaining data quality.

Genetic algorithms (GAs) are a powerful tool in portfolio management, optimizing asset selection and allocation by balancing multiple objectives like maximizing returns and minimizing risks. For example, a study on construction project portfolios used multi-objective GAs to align project selection with strategic goals and risk management, yielding optimized recommendations (Ghanbari et al., 2024). In finance, GAs have been employed to evaluate factors like mean, variance and skewness, improving investment outcomes (Mengzheng et al., 2024). Combining GAs with ML, such as deep learning, further enhances stock selection. A hybrid system integrating CNNs with GAs outperformed traditional methods in predicting stock trends and optimizing portfolios (Flores-Fernandez et al., 2022). Similarly, GAs paired with artificial neural networks refined investment strategies with high accuracy in stock performance predictions (Sangmin et al., 2020). GAs also incorporate risk models like Conditional Value at Risk, providing nuanced insights into asset interdependencies and risk exposure, as seen in portfolios including Bitcoin and gold (Bedoui et al., 2023). Despite their advantages, GAs face challenges like data quality requirements and risks of overfitting in complex models. Addressing these issues is essential to maximize their potential in real-world applications.

Deep learning approaches, particularly Recurrent Neural Networks (RNNs) and LSTM networks, are revolutionizing portfolio management by optimizing investment strategies. These models excel at capturing temporal dependencies in financial time series, enabling more accurate predictions of asset price movements (Chaher, 2023). Combining RNNs with CNNs further enhances stock selection and portfolio optimization, addressing the limitations of traditional methods (Mengzheng et al., 2024). Advanced techniques like the Task-Context Mutual Actor-Critic algorithm refine portfolio policies by integrating local asset features with global market context (Shantian, 2023). Expert-infused models add explainability and improve risk-adjusted returns, with studies reporting increases in Sharpe and Sortino ratios by up to 47.07% (Choi & Kim, 2024). Despite their advantages, deep learning models face challenges such as overfitting and interpretability, which can impact their robustness in volatile markets. Addressing these issues is critical for realizing their full potential in portfolio management.

Dynamic rebalancing strategies are crucial for optimizing asset allocation amid market fluctuations. These strategies leverage technical indicators, GAs, and behavioural models to improve portfolio performance. For instance, directional change methods, combined with GA, identify significant price movements and optimize portfolio weights, enabling rebalancing only during substantial changes, which reduces noise and enhances profitability (Almeida et al., 2023). Similarly, lag-optimized indicators like forecasted MACD and relative strength index (RSI) predict trend reversals, allowing for effective rebalancing during business cycle shifts and outperforming traditional approaches (Yeo et al., 2022).

Dynamic rebalancing also strengthens risk parity portfolios, such as the All-Weather portfolio. Techniques like optimal range rebalancing and trend-following based on moving-average distance improve risk-adjusted returns across diverse market conditions (Ning et al., 2023). Integrating investor behavior and market sentiment further refines these strategies, yielding higher returns per unit of risk compared to conventional methods (Wei et al., 2022). Despite their advantages, dynamic rebalancing introduces complexity, requiring careful implementation to avoid overfitting and ensure alignment with market conditions.

Portfolio management can significantly benefit from integrating Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO). PSO, known for its global optimization capabilities, effectively

navigates the solution space to identify optimal asset allocations, making it ideal for stock portfolio optimization (Thakkar & Chaudhari, 2021). Hybrid PSO methods that adaptively update parameters outperform traditional PSO in both computational efficiency and solution quality (Corazza et al., 2021). When combined with RL, PSO handles complex portfolio constraints effectively, achieving superior performance metrics like the Calmar ratio (Almahdi & Yang, 2019). Meanwhile, ACO excels in cardinality-constrained portfolio optimization, efficiently providing near-optimal solutions for NP-Complete problems (Can et al., 2023). Despite their advantages, both PSO and ACO face challenges, particularly in balancing exploration and exploitation to avoid local optima. Future research could focus on combining these methods to create even more robust solutions for portfolio management.

This study highlights the ability of DNNs to build prediction-based portfolio models, offering investors an innovative approach to portfolio optimization in the context of the Chinese stock market. Effective portfolio management governance enhances portfolio innovation and drives business performance, especially in terms of market and technological aspects. Key governance dimensions including formality, explicitness, information support and review frequency significantly influence portfolio innovativeness. Research has established a positive correlation between business performance and portfolio innovation, underscoring the pivotal role of innovation in organizational success (Urhahn & Spieth, 2014). The foundational principles and approaches in this field outline a trajectory that presents two distinct market analysis approaches. One approach starts with the efficient market approximation and lognormal price model, yielding outcomes such as MPT and Options Pricing Theory. The other approach begins by acknowledging market sentiment and evolving tools associated with technical analysis (Blakey, 2006).

PID controller optimization is a technique used in portfolio management strategies to achieve better risk-adjusted returns. Applying Model Predictive Control (MPC) to stock portfolio optimization yields simulation results, indicating a significant increase in investor capital. Moreover, it is feasible to employ a PID controller, which is commonly used in industrial processes, to manage financial market assets within a portfolio (Syaifudin & Putri, 2019). In addition, propose a method to predict future stock prices using the GBM-KF. Specific studies that have applied PID controllers in financial contexts, such as Syaifudin and Putri (2022), have demonstrated the effectiveness of MPC in portfolio optimization. This approach offers dynamic real-time adjustments to asset allocation an improvement over traditional static methods by enhancing portfolio stability and outperforming benchmarks through the use of a novel combination of unsupervised learning and control theory methods. A tactical asset allocation technique utilizes a PID controller to adjust portfolio asset weights to achieve the desired risk-adjusted return. By leveraging the PID control action, this approach is rooted in a model with a proven track record of success spanning over 50 years across diverse industries (Gandolfi et al., 2007). Portfolio optimization through deep RL integrates technical indicators and stock covariance. Additionally, employing the PID controller optimization technique for asset allocation in portfolio management proves practical and robust, as it does not require mathematical modeling of the process plant (Wang et al., 2022). In New Product Development portfolio management, control systems are viewed as heterogeneous actor networks that influence resource allocation, project interactions and the balance between centralized and decentralized controls (Donzello, 2014).

While the use of a PID controller in asset allocation, as referenced by Gandolfi et al. (2007), is less established in financial literature, our rationale for adopting this method lies in its ability to dynamically adjust portfolio weights based on market conditions. Unlike static optimization methods, the PID controller provides real-time responsiveness to volatility and momentum. It continuously fine-tunes asset allocations to minimize deviations from target risk-return profiles. Although not as widely recognized as traditional approaches like MVO, the PID controller's dynamic nature presents distinct advantages in volatile markets. That said, the method has limitations, including the need for careful tuning of the controller's parameters and the risk of overfitting in unpredictable markets. However, this study demonstrates its applicability in adaptive portfolio management, showing that it complements more established methods by adding real-time control and flexibility.

In finance management, EF involves selecting the best policies based on risk preferences and optimizing returns. Various approaches have been proposed to derive EFs. Three analytical derivations of the EF utilize various risk metrics to establish a closed-form solution to the relationship between expected returns and risk (Salas-Molina et al., 2023). DNN algorithms were utilized to compute the EF in portfolio optimization problems, demonstrating the efficiency of the methodology, particularly in high-

dimensional spaces (Warin, 2021). Data Envelopment Analysis is applied to measure hospital financial efficiency, while Stochastic Frontier Analysis identified contributing factors and strategies to enhance profitability (Wu, 2023). In portfolio construction, the EF is explored using both Mean-Variance and Black Litterman models, with findings indicating that the Black Litterman model exhibits lower risk (Subekti et al., 2022). Roll explored the differences in the EF geometry in mean-standard deviation and mean-variance spaces (Ababio et al., 2020; Bosch-Badia et al., 2017).

Studies such as those by Wen (2023) and Lu et al. (2018) emphasize that the static nature of traditional optimization models, particularly their reliance on historical data without adaptive capabilities, limits their efficacy in dynamic environments. Moreover, deep learning approaches for portfolio optimization (Ma et al., 2020) show that integrating ML can enhance decision-making by predicting future trends, but these models still often suffer from overfitting and high computational costs, lacking interpretability. Traditional techniques like MPT and its extensions, while foundational, assumes constant market conditions and normality in returns distributions, often failing to capture the complexities of nonlinear market dynamics and real-world asset behavior.

The proposed research fills a notable gap by integrating a PID controller into the optimization framework. This dynamic approach allows for real-time adjustments in asset weights, providing the adaptability that static models like MPT lack. Prior attempts to introduce adaptive strategies have been made, such as the use of MPC in portfolio optimization (Syaifudin & Putri, 2019), but they often focus on mathematical modelling without leveraging the strengths of unsupervised learning techniques like clustering. By incorporating momentum-based clustering, proposed study not only addresses the need for portfolio adaptability but also captures the inherent patterns in asset behavior, which enhances robustness against market volatility. This novel combination, as demonstrated in previous studies (G. Tang et al., 2022), allows for more granular, context-sensitive adjustments, positioning our approach as a significant advancement in adaptive portfolio optimization methodologies.

An innovative approach is proposed by integrating momentum clustering (Zhao et al., 2023). They introduced clustered federated learning based on momentum gradient descent (CFL-MGD) to address the federated learning problem of data heterogeneity. CFL-MGD partitions scattered clients into clusters based on their learning tasks and utilizes MGD to update the local model parameters. This study also presents a model and gradient averaging for global aggregation. The effectiveness of CFL-MGD is validated using the MNIST and CIFAR-10 datasets. An innovative approach introduces F2 CAN, a fast fuzzy clustering algorithm that utilizes a generalized momentum technique to expedite convergence (Azadeh et al., 2007; Hu et al., 2021). This demonstrates that the F2 CAN outperforms the existing FCAN algorithm in terms of effectiveness while maintaining the same degree of precision. The contributions of this study are including the following:

- Existing methods rely heavily on MPT and often fail to capture the complex, nonlinear dynamics of financial markets. Furthermore, they lack adaptability in effectively responding to changing market conditions. This study addresses several critical gaps in traditional portfolio optimization approaches.
- Introduce a novel data-driven framework that utilizes the strength of unsupervised ML techniques to uncover inherent patterns in stock behavior and market dynamics. By integrating momentum indicators and clustering algorithms, construct specialized portfolios tailored to distinct market profiles.
- Moreover, the innovative integration of control system principles, specifically the PID controller, into the optimization process allows for the dynamic adjustment of asset weights, enabling real-time course corrections and enhancing portfolio stability amidst market turbulence. This adaptive approach addresses a significant limitation of the traditional static optimization methods.

The remainder of this paper is organized as follows: [Section 2](#) describes the mathematical calculations and methodology. [Section 3](#) introduces the risk-adjusted performance metrics used for evaluating portfolio management. [Section 4](#) presents the data and explores dynamic portfolio optimization using clustering and PID control. [Sections 5 and 6](#) outline the integrated portfolio management strategy, along with the results and discussion of the proposed system. [Section 7](#) highlights aspects such as scalability, real-time adjustments and navigation through economic cycles. [Section 8](#) discusses the assumptions, limitations and potential for expanding the practical applicability of the proposed system. Finally, [Section 9](#) presents definitive conclusions and outlines future research directions.

2. Mathematical formulations

2.1. RSI

The pace and variation of price movements are measured using a momentum oscillator (RSI). It is calculated by using historical price data and formulas involving gains (increasing price movements) and losses (reducing price movements). The RSI formula normalizes the ratio of the average gain (AG) to average loss (AL) to a value between 0 and 100. Higher values indicate potentially overbought conditions, whereas lower values suggest oversold conditions (Wilder Jr., 1978).

The AG and AL can be calculated over a particular period (typically 14 periods), in which gains are considered positive changes in price and losses are considered negative changes. AG and AL can be calculated using Equations (1) and (2).

$$AG = \frac{\text{Sum of gains over the period}}{\text{Number of periods}} \quad (1)$$

$$AL = \frac{\text{Sum of losses over the period}}{\text{Number of periods}} \quad (2)$$

The relative strength and RSI are calculated using Equations (3) and (4).

$$RS = \frac{AG}{AL} \quad (3)$$

$$RSI = 100 - \frac{100}{1 + RS} \quad (4)$$

2.2. Average true range

Average True Range (ATR) is used to calculate asset volatility. It is determined by taking the average of the True Range (TR) during a certain period, where TR is equal to the greater of the following: the absolute value of the current high minus the previous close, the current low minus the previous close or the current high minus the current low (Wilder Jr., 1978). A rising ATR may indicate increasing volatility, whereas a falling ATR indicates decreasing volatility. TR is mathematically represented as Equation (5).

$$TR = \max (High-Low, |High - Previous Close|, |Low - Previous Close|) \quad (5)$$

The ATR is typically calculated as the average of the true range over a particular period. The average can be computed using various techniques, such as exponential moving average and simple moving average. The ATR can be calculated using Equation (6). Here, N is periods.

$$ATR = \frac{\sum_{i=1}^N TR_i}{N} \quad (6)$$

2.3. Dollar volume

The dollar volume measures the total value of the securities exchanged over a predetermined period. The computation involves multiplying the quantity of shares exchanged by share price. The dollar volume is calculated using Equation (7) (Achelis, 2000).

$$\text{Dollar volume} = \text{Shares Traded} \times \text{Price per Share} \quad (7)$$

2.4. Fama-French, rolling betas and percentage returns

The research calculates rolling factor betas for each stock through regression analysis using the Fama-French five-factor model, providing a measure of market responsiveness and sensitivity over time. Additionally, stock percentage returns are calculated for varying time lags from 1 to 12 months to capture performance over different investment horizons.

2.5. Momentum clustering

Clustering is a pivotal part of the methodology, categorizing stocks into homogeneous groups based on selected features that are indicative of market performance. This study employs K-means clustering, an approach for unsupervised ML that divides the dataset into 'K' clusters, each represented by a centroid (MacQueen, 1967).

The two stages of the K-means algorithm iteration are assignment and updating. Each data point is assigned to the closest centroid in the assignment stage using Euclidean distance, partitioning the data into K clusters. The mean of every point inside each cluster is used to calculate the new centroids during the update stage. The algorithm continues these steps until a stopping requirement is satisfied, usually when centroids stabilize or a maximum number of iterations is achieved.

Randomly select k initial cluster centroids. These centroids served as the initial cluster centers. The distance between each data point and centroid is determined and assigned to the nearest centroid. The most commonly used distance metric is the Euclidean distance. The cluster centroids can be calculated using Equation (8).

$$\sum_{i=1}^k \sum_{x \in S_i} \|x - \mu_i\|^2 \quad (8)$$

where S_i denotes the set of data points assigned to the i th centroid (μ_i).

The centroid of each cluster is recalculated by taking the mean of all data points that belong to that cluster. Mathematically, this step can be represented as follows: recalculation can be calculated using Equation (9).

$$\mu_i = \frac{1}{|S_i|} \sum_{x \in S_i} x \quad (9)$$

where $|S_i|$ represents the number of data points in cluster S_i .

K-means clustering is used to categorize the most liquid stocks based on their momentum profiles, employing RSI and ATR as technical indicators. RSI gauges the speed and change in price movements, whereas ATR captures price fluctuation intensity, providing insights into momentum and volatility. A fixed number of centroids within the K-means algorithm is used to establish a predetermined number of clusters. These preselected centroids act as reference points, guiding the grouping of stocks according to their momentum traits, as measured by RSI and ATR values, facilitating the creation of momentum tiers spanning from high to low.

2.6. PID control for portfolio optimization

PID is a widely used feedback control mechanism in engineering, particularly in systems in which continuous adjustments are required to maintain a desired setpoint (SP) or target. It is commonly employed in industrial processes, robotics, automotive control systems, etc. Here is a clear explanation of how PID control works (Aström & Hägglund, n.d.):

Proportional (P) Control: The P component of the PID control responds to the existing error between the measured process variable (PV) and the desired SP. In response to the current error, a control signal is generated in a directly proportional manner. This control signal $u(t)$ is given by Equation (10).

$$u(t) = K_p \cdot e(t) \quad (10)$$

where K_p is the proportional gain, which is a tuning parameter that determines the controller sensitivity. $e(t)$ is the error at time t and is calculated given (11).

$$e(t) = SP - PV \quad (11)$$

Essentially, the proportional-control component reduces the error by applying a control effort proportional to the current error. However, this may lead to a steady-state error because it does not consider the history of the error.

Integral (I) Control: The I component of the PID control aims to address the steady-state error by considering the accumulation of past errors over time. A control signal is generated based on the integral of the error over time. The integral control signal $u(t)$ is given by Equation (12).

$$u(t) = K_i \int_0^t e(\tau) d\tau \quad (12)$$

where K_i denotes the integral gain. The tuning parameter determines the response of the controller to accumulated errors. $\int_0^t e(\tau) d\tau$ represents the integral of the error from the initial to the current time t .

To eliminate steady-state error, the integral control element gradually modifies the control input in accordance with the total error. This facilitates the gradual guidance of the system towards the intended SP.

Derivative (D) Control: The derivative component of the PID control uses the error rate of change to predict future trends. This produces a control signal that is directly proportional to the error rate of change. Equation (13) provides the derivative control signal $u(t)$, is provided.

$$u(t) = K_d \frac{de(t)}{dt} \quad (13)$$

where K_d denotes the derivative gain. The adjustment parameter sets the weight assigned to the error rate of the change. $\frac{de(t)}{dt}$ represents the derivative of the error with respect to the time. It improves stability and reduces overshoot and oscillations by dampening the system response to rapid changes in error. In the PID controller, the control signal $u(t)$ is the sum of the proportional, integral and derivative control signals. The PID signal $u(t)$ is given by Equation (14).

$$u(t) = K_p \cdot e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \quad (14)$$

Proper tuning of the PID controller gains K_p, K_i and K_d is crucial for achieving the desired performance. This involves adjusting these parameters to balance the responsiveness, stability and robustness of the control system.

The PID controller is illustrated in Figure 1. In the feedback loop, the PID controller continuously monitors the PV and adjusts the control signal to minimize the error between the SP and actual PV. This forms a closed-loop feedback control system, where the controller continuously adjusts the control signal based on the feedback from the process. PID provides a flexible and effective method for controlling systems in various applications, offering a balance between responsiveness, stability and robustness through the proper tuning of its parameters.

2.7. EF

EF is used in investment to manage a portfolio to maximize the ratio of return to risk. The first step is to identify the asset classes or individual assets with unique risk and return characteristics. The portfolios are constructed by combining different assets in varying proportions, aiming for diversification to spread risk (Markowitz 1952).

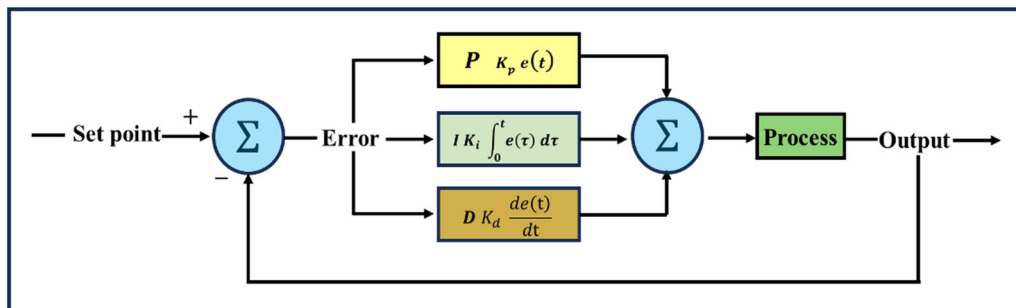


Figure 1. PID controller.

In portfolio theory, the weighted average of the expected returns of each asset in a portfolio is the expected return of the portfolio (μ). If we have a portfolio with n assets, each with expected returns $R_1, R_2, \dots, R_n$, the weights of the assets in the portfolio are $W_1, W_2, \dots, W_n$ respectively, and the expected return of the portfolio (μ_p) is given by Equation (15).

$$\mu_p = W_1R_1 + W_2R_2 + \dots + W_nR_n \quad (15)$$

The mathematical principles of the MPT underpin EF. The variance of a portfolio measures the dispersion of its returns and is calculated using the covariance matrix of asset returns and weights of assets.

Given that they provide the optimal risk-return trade-off, portfolios positioned on the EF are regarded as efficient. The Capital Market Line (μ), which is a tangent line drawn from the risk-free rate to the EF, shows the ideal risk-return trade-off for a risk-averse investor. A tangent line formed from the risk-free rate to EF is called. For a risk-averse investor, it is the best possible risk-return trade-off. The equation for the μ is given in Equation (16).

$$\mu = R_f + \frac{\mu_M - R_f}{\sigma_M} \cdot \sigma \quad (16)$$

$\mu, R_f, \mu_M, \sigma_M$ and σ are the expected return of the portfolio, risk-free rate, expected return of the market portfolio, standard deviation of the market portfolio and standard deviation of the portfolio, respectively.

EF aims to identify portfolios that reduce risk and maximize profits, or achieve a specific level of return with the lowest possible risk. This involves analyzing different portfolios along the EF and selecting the portfolio that aligns with the investor's risk appetite and return objectives.

Markets are dynamic and asset values fluctuate over time. To preserve the intended risk-return profile, it is imperative that the portfolio's performance be routinely monitored and adjusted as necessary. Rebalancing is the process of acquiring or disposing of assets to return a portfolio to its intended asset allocation.

Overall, EF provides investors with a systematic approach to constructing and managing portfolios that aim to maximize returns while effectively managing risk through diversification and optimizing asset allocation.

3. Risk-Adjusted performance measures for portfolio management

3.1. Rolling beta

Rolling beta (RB) (β_t) gauges an investment's systematic or market risk in relation to a benchmark (Chincarini & Kim, 2006). When beta is 1, it means that the investment moves in line with the benchmark; when beta is larger than 1, it indicates higher volatility; and when beta is less than 1, it shows reduced volatility. RB is a dynamic approach calculated over a rolling window of historical data, making it suitable for ongoing portfolio management. RB at time t is calculated using Equation (17).

$$\beta_t = \frac{\sum_{i=t-T+1}^t (R_{pt} - \bar{R}_m) \cdot (R_{mt} - \bar{R}_m)}{\sum_{i=t-T+1}^t (R_{mt} - \bar{R}_m)^2} \quad (17)$$

where R_{pt} and R_{mt} represent the benchmark and portfolio returns at time t , respectively, T represents the window size (number of previous periods), and $\bar{R}_m$ is the benchmark's average return over window T .

3.2. Rolling Sharpe Ratio (RSR)

Excess Returns (ER) earned per unit of risk is assessed using a risk-adjusted performance metric (RSR). A rolling method is employed by dividing the average ER over a window by the standard deviation of portfolio returns within that window. Compared to periods with lower ratios, a higher RSR signifies stronger risk-adjusted returns. The RSR (SR_t) at time t is calculated using Equation (18) (Sharpe 1994).

$$SR_t = \frac{\left(\frac{\sum R_{pt} R_f}{T}\right)^{\frac{1}{2}}}{\sigma_t} \quad (18)$$

where σ_t is the standard deviation of the portfolio returns within window T and R_f is the risk-free rate at time t .

3.3. Cumulative returns

Cumulative returns (CR) calculates the portfolio's overall growth over a given timeframe, reflecting the compounded effect of individual returns (DeBondt & Thaler, n.d.). This provides a clear picture of the portfolio's overall wealth growth trajectory. CR (CR_t) at time t is calculated using Equation (19).

$$CR_t = \left(\prod (1 + R_{pt}) - 1 \right) \times 100 \quad (19)$$

3.4. Beta (β)

In comparison to a predetermined benchmark, beta gauges a portfolio's systematic or market risk. Close synchronization with market movements is indicated by a high beta, which can lead to better returns but also larger losses during downturns. Equation is used to compute the β_t at time t (20).

$$\beta_t = \frac{\sum_{i=t}^T \sum_{j=1}^T (R_{pt} - \bar{R}_m) \cdot (R_{mt} - \bar{R}_m)}{\sum_{i=t}^T \sum_{j=1}^T (R_{mt} - \bar{R}_m)^2} \quad (20)$$

where R_{mt} , $\bar{R}_m$ and T are the market returns at time t , average market return and total number of observations, respectively.

3.5. Alpha (α)

The excess return that a portfolio generates over what is predicted by its beta is known as the alpha. A positive alpha suggests that the manager's skill or active management strategy adds value, while a negative alpha implies underperformance. The α can be calculated using Equation (21). Here, $\bar{R}_p$ is the average return on the asset/portfolio (Sharpe, 1994).

$$\alpha = \bar{R}_p - (R_f + \beta \cdot (\bar{R}_m - R_f)) \quad (21)$$

3.6. R-Squared (R^2)

The R^2 measures how well the benchmark return explains the variations in portfolio returns. A higher R^2 suggests a stronger relationship between the portfolio and benchmark. The R^2 can be calculated using Equation (22). Here, $\hat{R}_p$ represents the asset or the portfolio's expected return.

$$R^2 = 1 - \frac{\sum (R_{pt} - \hat{R}_p)^2}{\sum (R_{pt} - R_p)^2} \quad (22)$$

3.7. Mean Annual Return

The average annualized return of a portfolio over a given period is known as Mean Annual Return (MAR), which provides a measure of the portfolio's overall growth trajectory. The MAR can be calculated using Equation (23) (Malkiel, 2015).

$$MAR = \left((1 + R_1) \cdot (1 + R_2) \cdot \dots \cdot (1 + R_n)^{\frac{1}{n}} \right) - 1 \quad (23)$$

where $R_1, R_2, \dots, R_n$ are each period's investments, and n is the total number of periods.

3.8. ER

The return that a portfolio generates above the risk-free rate is represented by the ER, aiding in the evaluation of the portfolio's capacity to reimburse investors for their market risk, where the risk-free rate of return (R_f) and average RoI ($\bar{R}_p$) are determined. The ER can be computed using Equation (24).

$$\text{Excess Return} = \bar{R}_p - R_f \quad (24)$$

3.9. Standard deviation (σ)

This quantifies how widely distributed or volatile a portfolio's returns are around its mean, offering vital risk assessment. The σ can be calculated using Equation (25) (Elton et al., 2014).

$$\sigma = \sqrt{\frac{\sum (R_{pt} - \hat{R}_p)^2}{n - 1}} \quad (25)$$

3.10. SR

The ER earned per unit of risk is determined by SR, which is determined by the portfolio return standard deviation. Better risk-adjusted performance is indicated by a larger ratio. SR can be computed using Equation (26).

$$SR = \frac{(\hat{R}_p - R_f)}{\sigma} \quad (26)$$

3.11. Treynor ratio

The ER obtained per unit of systematic risk is the main emphasis of Treynor ratio (TR), as determined by the beta of the portfolio. Comparing portfolios with different degrees of market exposure would be helpful. TR can be calculated using Equation (27).

$$\text{Treynor ratio} = \frac{\bar{R}_p - R_f}{\beta} \quad (27)$$

3.12. Sortino ratio

The ER is determined by the Sortino ratio (SRR) for each unit of downside risk as determined by the negative return standard deviation (downside deviation). It is valuable for investors to adverse downside risks. The SRR can be calculated using (28) (Elton et al., 2014).

$$\text{Sortino ratio} = \frac{\bar{R}_p - R_f}{\sqrt{\frac{\sum (\text{Downside deviation})^2}{n-1}}} \quad (28)$$

3.13. Jensen's alpha

The difference between the portfolio's actual return and its anticipated return, as determined by the Capital Asset Pricing Model, is measured by the Jensen's alpha (JA). A positive alpha denotes superior performance and highlights the manager's aptitude or successful tactics. JA can be calculated using Equation (28).

$$\text{Jensen's alpha} = \bar{R}_p - (R_f + \beta \cdot (\bar{R}_m - R_f)) \quad (29)$$

4. Data and dynamic portfolio optimal via clustering and PID control

The data used in this study is obtained from Yahoo Finance, a free and reliable financial data platform. Yahoo Finance provides open-source data accessible through its APIs, and the data collected is intended for research and educational purposes. The nature of the data includes daily stock market information which are Open, High, Low, and Close (OHLC) prices. The data spans a period of 8 years, ending in October 2023. The dataset comprises information from the Companies Listed in the S&P 500 Index, with four parameters (OHLC) recorded on a daily timeframe. The data is accessed using the *yfinance* package, which utilizes Yahoo's publicly available APIs. This package is open-source and widely used for data retrieval and analysis in financial research. No special rights or permissions are required to obtain this data as it is publicly available and provided by Yahoo Finance for research and educational purposes.

To ensure the integrity of our analysis, we implemented a rigorous data preprocessing protocol. We addressed minor cases of missing data through interpolation methods, ensuring that the dataset remained robust for analysis. This step is crucial as it helps maintain the continuity of time series data, which is essential for accurate performance evaluation. Calculated various technical indicators, such as the Relative RSI and ATR, which serve as momentum indicators. These features were derived from the raw OHLC data to enhance our analysis and support the momentum clustering process. To facilitate the clustering process and ensure that all features contributed equally to the analysis, we normalized the data. This step helps in mitigating the impact of outliers and ensures that the clustering algorithm operates effectively. We divided the dataset into different time intervals (one, two, three, six, nine, and twelve months) to evaluate the performance of the portfolio over various horizons. This segmentation allows for a comprehensive analysis of the strategy's effectiveness across different market conditions.

To identify the most relevant variables for clustering, we began with an exploratory analysis to evaluate how different variables impact the differentiation of portfolio characteristics. Key financial indicators, such as volatility, SR and sector allocation, were prioritized for their relevance to typical risk-return profiles in portfolio construction. Variables like RSI, ATR and dollar volume were included due to their established effectiveness as indicators of momentum and volatility in financial studies.

To tune the PID controller parameters, a grid search method is employed to optimize the proportional, integral and derivative gains, focusing on achieving both stability and responsiveness. For the k-means clustering algorithm, the elbow method is applied to determine the optimal number of clusters, balancing simplicity and accuracy. The PID controller's gains (K_p , K_i , K_d) were then manually refined to ensure portfolio asset weights adjusted responsively while maintaining stability.

The key objective of using the PID controller is to manage the portfolio's risk-return trade-off by continually adjusting the weights based on real-time data, while maintaining performance stability and responsiveness. The PID method is applied to adjust the weights of the 155 stocks in the pool, optimizing for risk and return based on their individual characteristics. The PID controller dynamically adjusts the portfolio's composition in response to market changes, balancing responsiveness with stability. Despite the large stock pool, the PID method efficiently handles weight adjustments, focusing on minimizing risk and maximizing returns rather than directly selecting or forecasting individual stock prices.

The combined method operates as follows: First, the portfolio's assets are selected, and their initial weights are assigned using traditional portfolio optimization techniques, such as equal weights optimization. Subsequently, the PID controller adjusts these weights in real time. The proportional term (P) focuses on the current error, or the deviation from the desired portfolio allocation. The integral term (I) accumulates past errors to correct for cumulative misalignments, and the derivative term (D) anticipates future changes by considering the rate of change of the error. This continuous adjustment process enables dynamic rebalancing, making the strategy more adaptive and responsive than static approaches.

5. Overall integrated portfolio management strategy

A schematic of the proposed integrated portfolio-management strategy is shown in [Figure 2](#). The process begins by calculating the essential technical indicators for all the instruments. These indicators include RSI, ATR, and dollar volumes. The RSI measures the speed and magnitude of price movements, while ATR captures price volatility. Dollar Volume quantifies the total value of securities traded over a

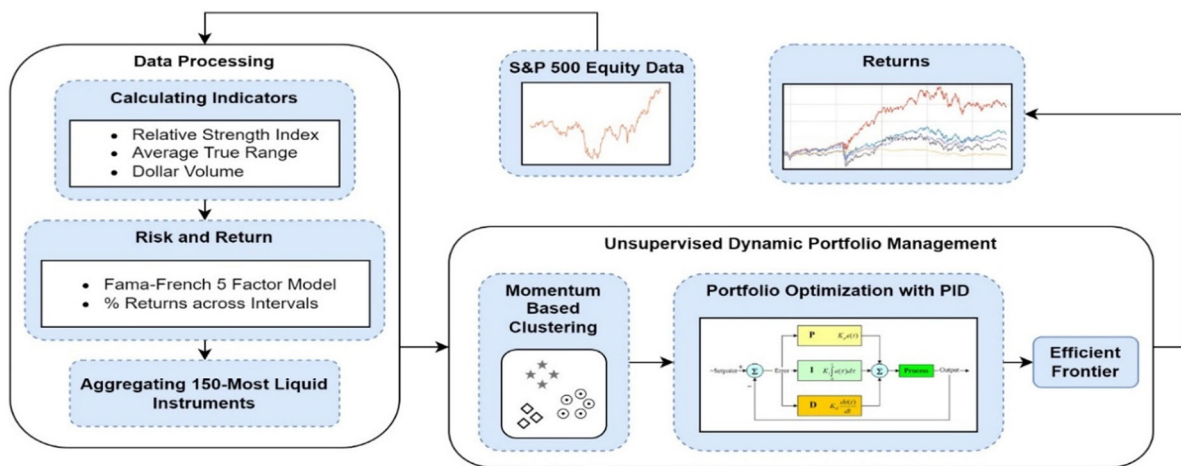


Figure 2. Schematic efficient wealth management strategies.

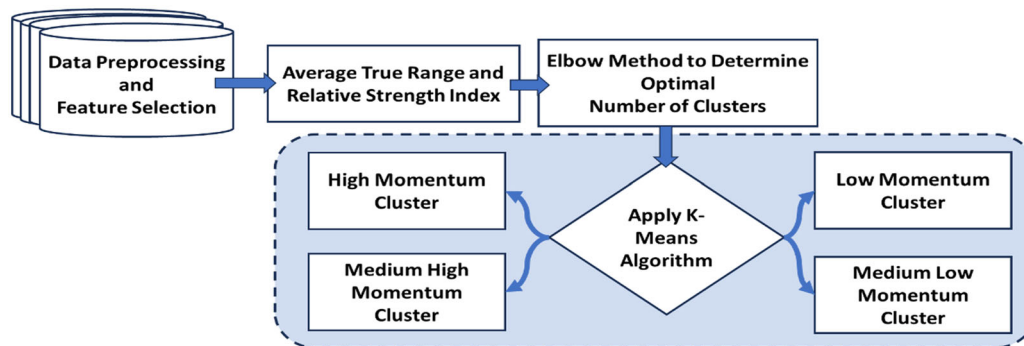


Figure 3. Schematic of clustering process.

specific period and is used to aggregate approximately the top 150 most liquid stocks. The Fama-French Five Factor model is used to calculate the rolling factor betas for each stock through regression analysis. These RBs provide measures of market responsiveness and sensitivity over time. Additionally, the stocks' percentage returns are calculated for different time lags, ranging from 1 to 12 months, to capture performance over diverse investment horizons.

The pivotal step in the methodology is momentum clustering, where the K-means clustering algorithm is employed to group stocks into homogeneous clusters based on their momentum profiles, as indicated by the RSI and ATR (Hruschka & Natter, 1999). This unsupervised ML technique partitions the data into 'K' clusters, each represented by a centroid, facilitating the construction of specialized portfolios tailored to distinct market profiles based on levels of momentum.

Subsequently, an EF analysis is employed to construct portfolios with optimal risk-return trade-offs. This identifies asset combinations that yield optimal returns while effectively managing risk through diversification and optimizing asset allocation. Here, the step involves identifying individual assets or asset classes with unique risk and return characteristics, combining them in varying proportions and selecting the portfolios positioned on the EF, which represent the optimal risk-return trade-offs.

Figure 3 presents a flowchart outlining the process of clustering S&P 500 stocks into distinct groups high, medium, and low momentum on a monthly basis, using RSI and ATR as key features. The clustering process begins with preparing raw S&P 500 stock data by cleaning and organizing it. Key features, such as ATR and RSI, are selected to analyze stock momentum. ATR measures how much stock prices fluctuate, providing an idea of market volatility, while RSI tracks the speed and magnitude of price changes, reflecting momentum strength. To determine the right number of clusters, the Elbow Method is used, where a graph compares different numbers of clusters, and the 'elbow point' indicates the optimal choice. Using this information, the K-means algorithm groups stocks based on their ATR and RSI values into four clusters: High Momentum (stocks with strong upward movement), Medium-High

Momentum (stocks with moderately strong movement), Medium-Low Momentum (stocks with slightly weaker movement) and Low Momentum (stocks with weak or downward movement). A schematic diagram can illustrate the entire process, from data preparation and feature selection to the Elbow Method and K-means clustering, providing a clear understanding of how these steps contribute to optimizing a stock portfolio.

6. Result and discussion

A decade of open-source S&P 500 stock data is utilized for analysis. The strategy begins by calculating technical indicators and the Fama-French model based rolling factor betas for stocks, followed by momentum clustering using the K-means algorithm to group stocks based on their momentum profiles. Employing a rigorous data processing protocol, calculated essential indicators, including the RSI, ATR and Dollar Volume. These metrics serve as both momentum indicators and aids in aggregating most liquid stocks for analysis. After computing the indicators, evaluate the Fama-French five factors, which serve as constants in calculating the Rolling Factor Betas through regression analysis. Compute the percentage returns for the following intervals: one, two, three, six, nine, and twelve months. K-means clustering with RSI and ATR indicators successfully segments the most liquid stocks into clusters with four predetermined centroids ranging from High to Low momentum, as shown in Figure 4. Here, the focus is on investigating stocks within a high-momentum cluster.

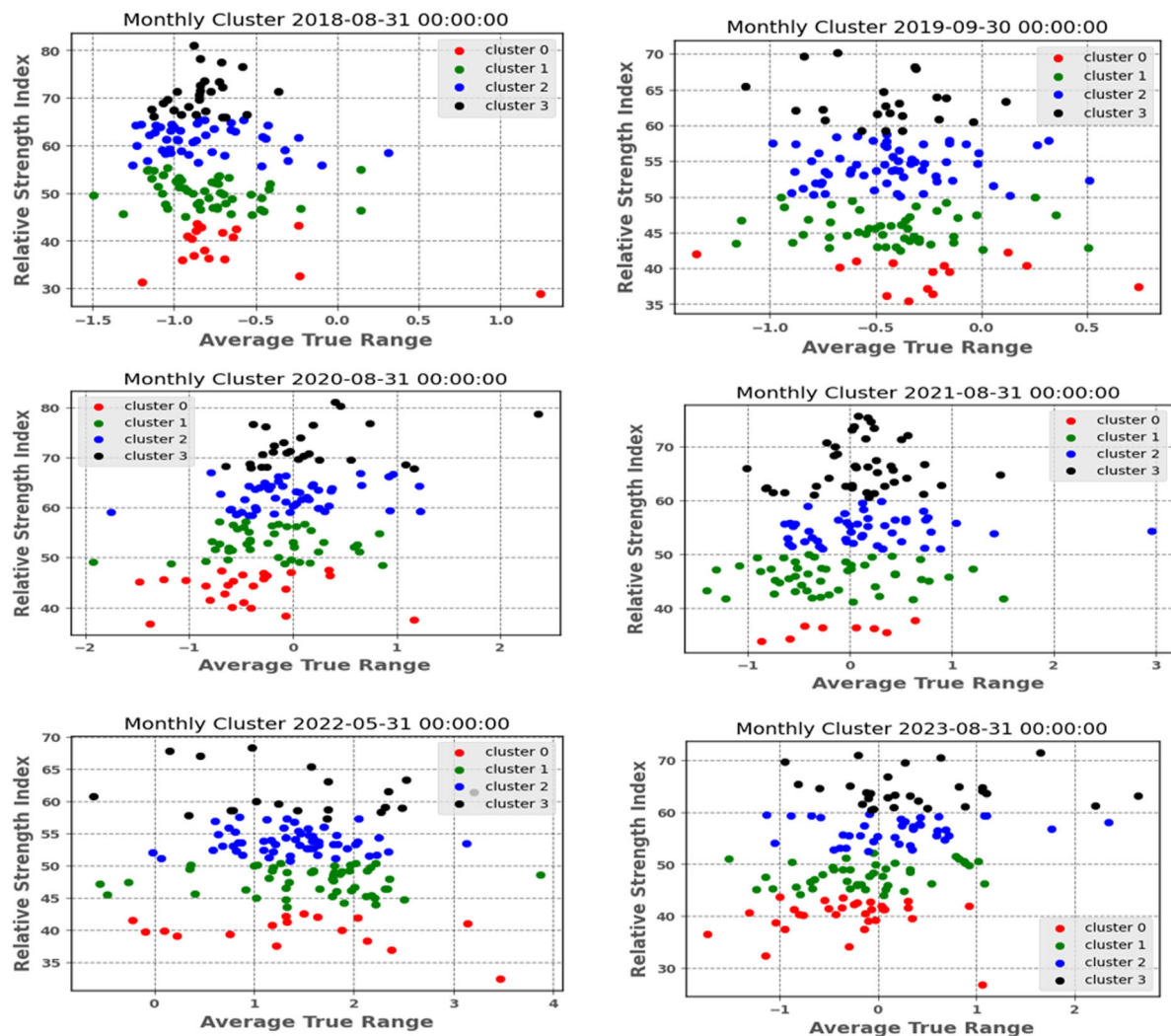


Figure 4. Sample momentum clustering.

PID controller to determine the optimal weights for each component within the cluster. Building on this approach, utilize an EF model to identify stock combinations that yield optimal returns. Table 1 presents the list of selected stocks comprising the final portfolio upon which both optimization and EF algorithms is applied. While the PID controller is traditionally used in control systems, demonstrate its adaptability in finance by dynamically adjusting asset weights in response to market fluctuations, similar to how it corrects errors in control systems. This dynamic adjustment sets it apart from static methods like Markowitz's MPT, enabling more responsive portfolio management. To compare, implemented the process without the PID controller, yielding returns of 104.92%. However, with the PID controller, returns increased to 135%, demonstrating its superior ability to capture market shifts and optimize returns. The clustering method complements the PID by grouping assets based on momentum, facilitating more precise, real-time adjustments. This creates a seamless connection between asset clustering and portfolio optimization, enhancing overall performance.

Table 2 shown the yearly RB comparison against benchmarks. The RB analysis provides valuable insights into the performance of various benchmarks compared to the portfolio strategy over the years. Across the board, the S&P 500 consistently exhibits a strong positive correlation with the portfolio strategy, with RBs ranging from 0.54247 in 2021 to 0.90568 in 2019. This indicates that movements in the S&P 500 closely track those of the portfolio strategy, suggesting a significant influence of large-cap U.S. stocks on the portfolio's performance.

In contrast, DJIA and Russell 2000 show slightly lower but still positive RBs compared to the portfolio strategy. The RBs of DJIA range from 0.48730 in 2021 to 0.86949 in 2019, while Russell 2000 exhibited RBs ranging from 0.78462 in 2021 to 0.99277 in 2019. These findings suggest that both the DJIA and Russell 2000 have a positive relationship with portfolio strategy, albeit with some variability over time.

Moving beyond domestic indices, the Barclays US bonds present a mixed picture of their correlation with portfolio strategy. Although the RBs for Barclays US bonds consistently show negative values, ranging from -0.02861 in 2018 to -0.01405 in 2021, they indicate a weak inverse relationship with portfolio strategy. This implies that changes in the bond market may not have had a significant effect on the portfolio's overall performance over the study period.

Examining international benchmarks, both the DAX and FTSE demonstrate positive RBs with portfolio strategy, albeit with some fluctuations. Figure 5 shows the RB comparison against the daily benchmarks. The DAX exhibits RBs ranging from 0.28916 in 2021 to 0.44030 in 2023, indicating a moderate positive correlation with portfolio strategy. Similarly, the FTSE shows RBs ranging from 0.18021 in 2018 to 0.50770 in 2019, highlighting its positive relationship with portfolio strategy, albeit with some variability

Table 1. List of instruments comprise the portfolio.

INTC	AIG	SLB	NVDA	HUM	PFE	GD	KR	ELV	CMCSA	CCL
JPM	UNH	MA	REGN	DLTR	RCL	ADBE	WFC	AMD	WYNN	GILD
FIS	BDX	SBUX	GS	SCHW	MET	BA	WBA	JNJ	ISRG	BSX
QCOM	AMAT	NKE	NFLX	MCD	DHR	AAPL	UPS	ADP	GE	LRCX
CHTR	VZ	AMGN	MS	LULU	LMT	CMG	MCK	MO	HON	BAC
TMUS	GOOG	MDT	FCX	AMT	KO	META	WDC	TSLA	XOM	
EA	DAL	ULTA	UNP	GM	ABBV	WMT	EBAY	MSFT	MRK	
STZ	TMO	EOG	AZO	TJX	HD	USB	BMJ	NOW	MAR	
LOW	PG	NXPI	AAL	CTSH	ABT	ADSK	EXPE	LUV	CSX	
PANW	ORLY	CI	UAL	DE	BIIB	VLO	MRNA	MMM	FDX	
TXN	AVGO	T	DIS	VRTX	PXD	RTX	PNC	V	PM	
LIN	CRM	UBER	COST	CL	NEM	DD	INTU	MDLZ	TGT	
CSCO	ORCL	OXY	BKNG	IBM	F	CVX	COP	DG	CVS	
ACN	ILMN	NEE	MPC	CAT	HAL	MU	AMZN	C	PARA	
LLY	FI	GOOGL	BLK	SHW	PEP	AXP	PYPL	BRK-B	NOC	

Table 2. Yearly rolling beta comparison against benchmarks.

Year	S&P 500	DJIA	Russell 2000	Barclays US bonds	DAX	NASDAQ	FTSE	Strategy returns
2018	0.80229	0.80267	0.79313	-0.02861	0.32025	0.95460	0.18021	1.00000
2019	0.90568	0.86949	0.99277	-0.06139	0.63322	1.07401	0.50770	1.00000
2020	0.74475	0.76241	0.88438	0.00971	0.43000	0.79807	0.39492	1.00000
2021	0.54247	0.48730	0.78462	-0.01405	0.28916	0.63415	0.21903	1.00000
2022	0.77909	0.64033	0.87416	0.07124	0.38665	0.93081	0.25586	1.00000
2023	0.75498	0.61651	0.85914	0.08800	0.44030	0.91106	0.26791	1.00000

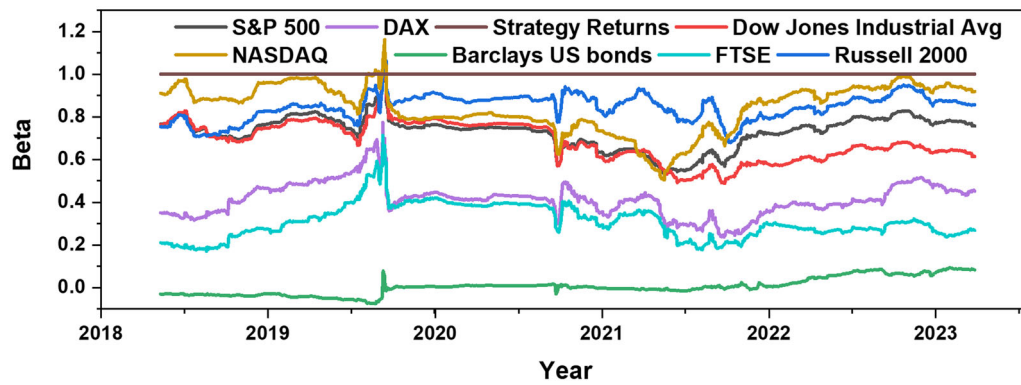


Figure 5. Rolling beta comparison against benchmarks for daily.

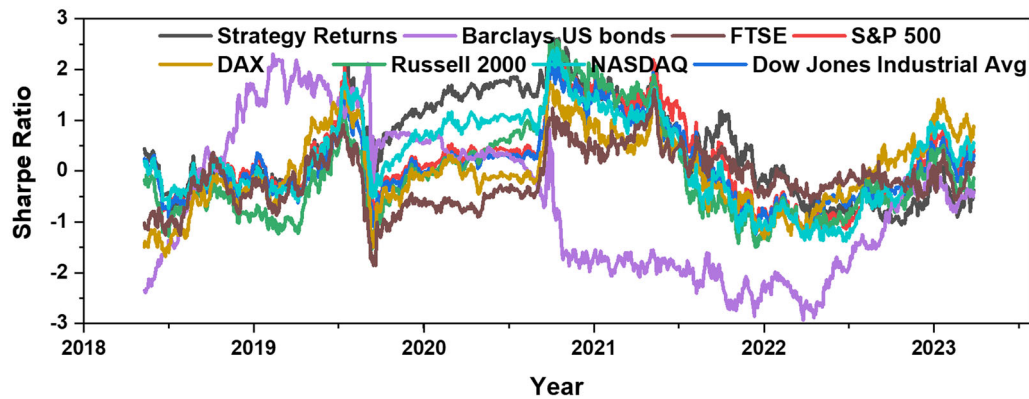


Figure 6. RSR comparison against benchmarks.

Table 3. Yearly RSR comparison against benchmarks.

Year	Strategy returns	S&P 500	DJIA	Russell 2000	Barclays US bonds	DAX	NASDAQ	FTSE
2018	-0.52621	-0.75042	-0.74078	-1.17971	-1.54657	-1.52467	-0.55989	-1.04501
2019	0.96705	0.88440	0.51486	0.30420	1.28613	1.13081	0.72579	0.49702
2020	1.84822	0.39957	0.31095	0.60681	0.32269	-0.13036	1.05709	-0.39079
2021	0.95451	1.26045	0.82663	0.44630	-1.88922	0.47446	0.73215	0.45014
2022	-0.53326	-1.00490	-0.61525	-0.83606	-1.98825	-0.55601	-1.21512	-0.20019
2023	-0.38430	0.42067	0.31425	-0.13634	-0.40883	0.88145	0.55805	0.16463

over time. Overall, these results highlight the importance of considering both the local and global market changes when evaluating portfolio performance.

The RSR comparison with the benchmarks is shown in Figure 6. Table 3 presents a yearly RSR comparison against the benchmarks. In contrast to RBs, which gauge the relationship between assets, SR considers volatility and returns, offering a more thorough evaluation of investment performance. Across the analysed period, the strategy returns exhibit varying levels of risk-adjusted performance compared with the benchmarks.

In 2018, the strategy returns showed a negative SR, meaning that the portfolio's returns are insufficient to offset the degree of risk. This contrasts with some benchmarks, such as the S&P 500 and DJIA, which also experienced negative SR, suggesting widespread challenges in achieving favourable risk-adjusted returns amid market volatility. Conversely, assets such as Barclays US bonds demonstrated an even lower SR, reflecting relatively poor risk-adjusted performance compared to the portfolio strategy.

In 2019, strategy returns exhibited a significantly improved SR, reflecting a period of robust risk-adjusted performance. Notably, the strategy returns outperformed several benchmarks, including the S&P 500 and DJIA, in terms of risk-adjusted returns. This suggests that the portfolio's allocation and investment decisions may have been particularly effective in generating superior risk-adjusted returns compared with these major market indices.

However, in subsequent years, the strategy returns experienced fluctuations in risk-adjusted performance. While 2020 saw a notable increase in the strategy's SR, indicating strong risk-adjusted returns compared with most benchmarks, subsequent years witnessed a decline in performance. In 2021 and 2022, the strategy returns exhibited a negative SR, suggesting a period of underperformance relative to the benchmarks analysed.

Metric comparisons against benchmarks are presented in Table 4. MAR, ER and standard deviation shed light on a strategy's performance in terms of returns and volatility. While the strategy demonstrates an MAR of 0.17343 and an ER of 0.12464, which is the highest in both cases across benchmarks, most benchmarks, such as Barclays US bonds, exhibit lower mean returns and negative excess returns, suggesting inferior performance relative to the strategy. Additionally, the strategy's standard deviation of 0.21566 reflects its level of volatility compared to the benchmarks.

The ER produced per unit of risk (standard deviation) is assessed by SR, a commonly used indicator of risk-adjusted returns. The strategy's positive SR of 0.57794 suggests that it offers a favourable trade-off between risk and return when compared to some benchmarks, such as the DJIA and S&P 500, which have lower SR. The strategy's ability to outperform even in volatile conditions, such as in 2020, reflects its adaptability in rapidly changing market environments. The robust RB values also indicate positive correlations with major indices, confirming the strategy's alignment with broader market trends while offering superior risk-adjusted returns.

Further analysis shows that the higher Sharpe and Sortino ratios demonstrate better risk management, especially in 2020, when compared to benchmarks like the S&P 500. The negative SR in 2022 hints at underperformance during specific market conditions, which could be attributed to global market volatility during that period. This suggests that refining the model to account for extreme conditions may further enhance its robustness. However, the strategy's overall resilience across different years supports its potential as a reliable portfolio management tool in varying financial climates.

Figures 7–9 show the comparison of yearly returns, yearly RB and performance metrics, respectively. The yearly returns comparison display highlights the strategy's annual returns in comparison to the S&P 500, DJIA, Russell 2000 and NASDAQ. The strategy consistently delivers higher returns than these benchmarks. The Yearly Rolling Beta Comparison Chart showcases the strategy's annual RB relative to major benchmarks. Notably, the strategy maintains a stable beta of 1.0, indicating it closely aligns with its own unique risk profile, while the benchmarks exhibit varying beta levels. The Performance Metrics Comparison Chart provides a bar chart analysis of key metrics, including the SR, Sortino Ratio, Jensen's Alpha and more, comparing the strategy to the benchmarks. The strategy excels in risk-adjusted performance (as shown by superior Sharpe and Sortino Ratios) and other metrics, underscoring its strong overall performance.

To evaluate the robustness of the proposed strategy against traditional benchmarks, we applied two statistical tests on the SR: The Paired *t*-Test and the Wilcoxon Signed-Rank Test. These tests rigorously determine whether performance differences are statistically significant or due to chance. The Paired *t*-Test, assuming normally distributed differences, compared the mean SRs of the strategy and benchmarks, yielding a *p* value of .0018. This is well below the conventional 0.05 threshold, confirming that the strategy significantly outperforms benchmarks in risk-adjusted returns.

To account for non-normality, the Wilcoxon Signed-Rank Test was also performed, assessing whether median differences were systematically positive. It produced a *p* value of .0156, further supporting the statistical significance of the strategy's superior performance. The consistency of results across both tests

Table 4. Metrics comparison against benchmarks.

Metric	Strategy returns	S&P 500	DJIA	Russell 2000	Barclays US bonds	DAX	NASDAQ	FTSE
Overall beta	1.00000	0.74489	0.70420	0.86277	0.01424	0.41353	0.85144	0.32816
Overall alpha	0.00000	−0.00016	−0.00021	−0.00038	0.00003	−0.00017	−0.00011	−0.00018
R-squared	1.00000	0.61114	0.54522	0.51419	0.00272	0.18924	0.55878	0.17135
Mean annual return	0.17343	0.08840	0.06998	0.05300	0.01040	0.02814	0.12111	0.01250
Excess return	0.12464	0.03960	0.02119	0.00421	−0.03840	−0.02065	0.07232	−0.03629
Standard deviation	0.21566	0.20549	0.20568	0.25948	0.05892	0.20501	0.24564	0.17097
Sharpe ratio	0.57794	0.19271	0.10301	0.01621	−0.65168	−0.10073	0.29441	−0.21228
Treynor ratio	0.12464	0.05316	0.03009	0.00487	−2.69630	−0.04994	0.08494	−0.11060
Sortino ratio	0.83780	0.26125	0.14144	0.02217	−0.88201	−0.13818	0.40771	−0.28321
Jensen's alpha	0.00121	−0.05234	−0.06574	−0.10229	−0.04015	−0.07170	−0.03278	−0.07680

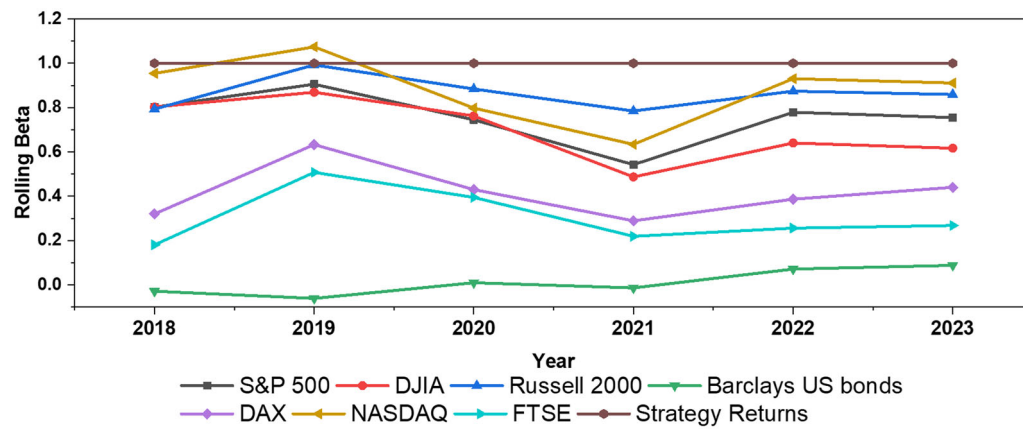


Figure 7. Yearly returns comparison.

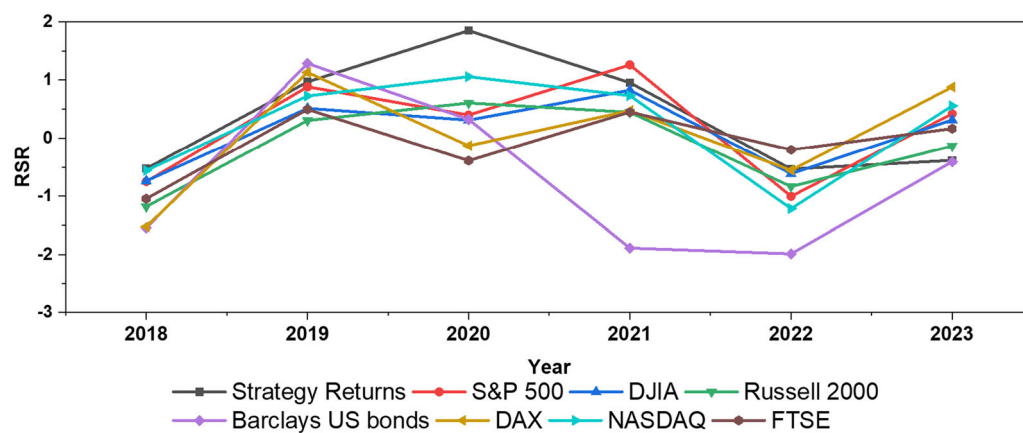


Figure 8. Yearly RSR comparison against benchmarks.

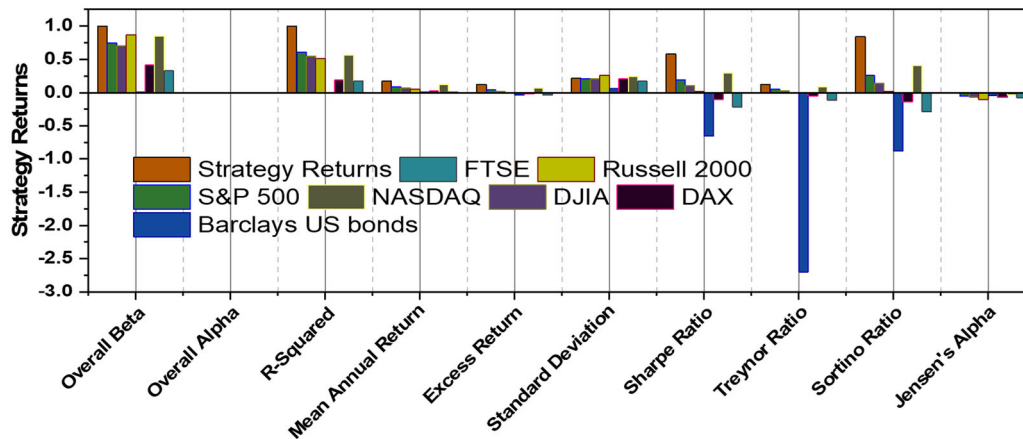


Figure 9. Metrics comparison against benchmarks.

highlights the strategy's effectiveness, with low p values indicating that observed improvements in risk-adjusted returns are unlikely due to random variation.

With a positive Treynor Ratio of 0.12464, the strategy indicates an ER relative to its systematic risk, suggesting a potentially efficient allocation of assets to generate returns compared to others. A variation of the SR that concentrates on downside risk is the Sortino Ratio, which assesses the ER per unit of downside risk (negative standard deviation). The approach shows favourable risk-adjusted returns in relation to downside risk, with a positive Sortino Ratio of 0.83780. This is in contrast to several benchmarks such as the DJIA and S&P 500, which have substantially lower Sortino Ratios. Based on systematic risk

exposure, the strategy shows a modestly positive ER in relation to its projected returns, with a positive Jensen's alpha of 0.00121. Figure 10 shows the daily returns comparison between the strategy returns of S&P 500 and SPY Buy and Hold.

The histogram of yearly returns presented in the Figure 11. The histogram visualizes the distribution of yearly returns for the strategy and benchmarks, highlighting variability and performance trends. The strategy's returns demonstrate a higher concentration in positive ranges compared to benchmarks like the S&P 500 and Russell 2000, which show broader distributions. Negative returns are more frequent for some benchmarks, such as Barclays US Bonds, reflecting greater downside risk. This analysis emphasizes the strategy's consistency and reduced exposure to extreme losses.

Figure 12 depicts CR for four strategies: Inverse Volatility, Strategy Returns, Minimum Variance and SPY Buy & Hold from 2017 to 2024. Strategy Returns and Minimum Variance exhibit higher cumulative

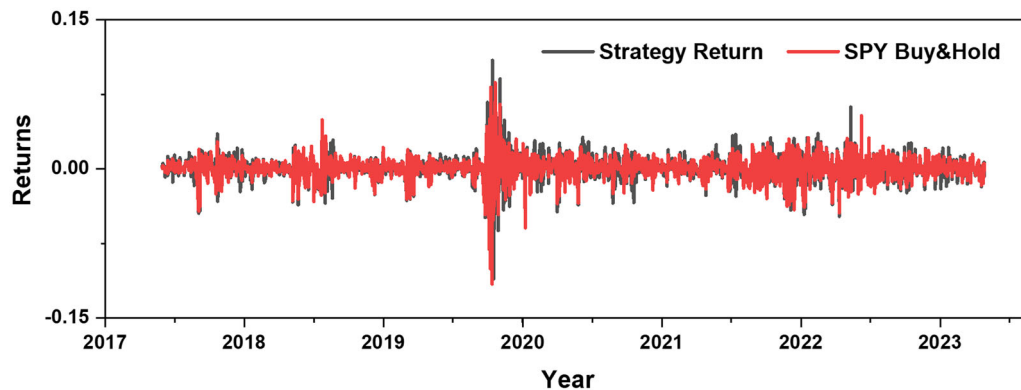


Figure 10. Daily returns comparison against S&P 500.

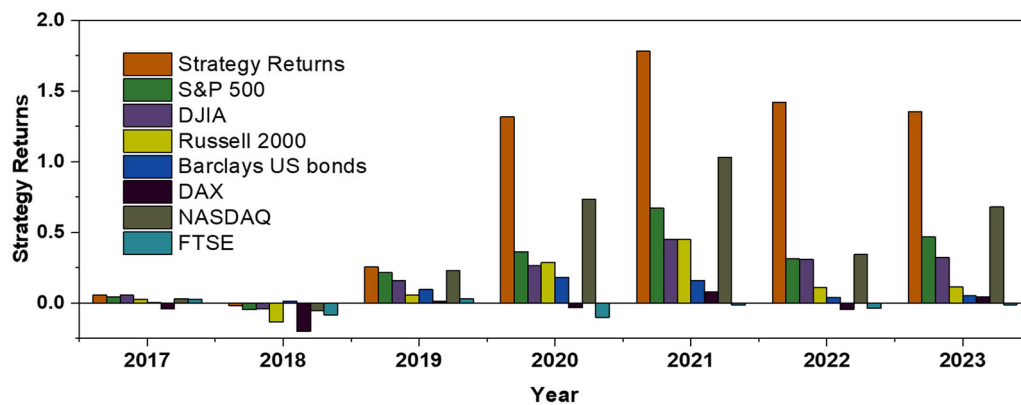


Figure 11. Histogram of yearly returns.

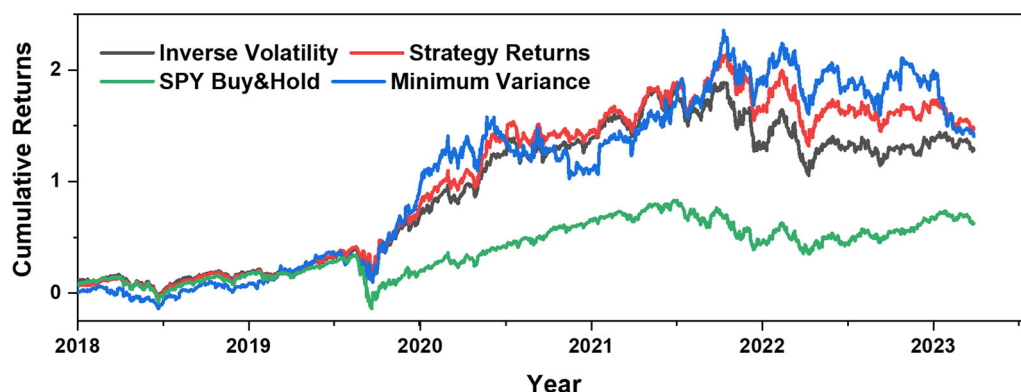


Figure 12. Strategy returns Vs alternate portfolio methods.

growth, outperforming SPY Buy & Hold and Inverse Volatility, especially during market uptrends. Inverse Volatility remains more stable but lags in performance, indicating lower risk exposure. The results suggest Strategy Returns balances growth potential and volatility more effectively compared to benchmarks.

Figure 13 shows the comparison of the compounding returns against benchmarks. A year-wise return comparison against benchmarks is given in Table 5. Across each year of analysis, the strategy consistently outperformed the benchmarks in terms of the returns. In 2017, the strategy delivered a return of 5.45%, surpassing the returns of all benchmarks albeit marginally. Notably, in 2018, a challenging year for many investors, the strategy mitigated losses more effectively than the benchmarks, with a return of -1.66% compared to the larger downturns experienced by the S&P 500 (-4.70%) and DJIA (-4.00%).

The statistical analysis reveals that for daily returns, the strategy shows a moderate but statistically significant difference from zero ($t = 1.93$, $p = .044$). When compared to the S&P 500 CR, the paired t -test ($t = 1.46$, $p = .045$) indicates a significant difference in the overall growth trajectory between the strategy and the benchmark. This difference is likely due to variations in return distribution, where the strategy captures more momentum-driven gains or exhibits different risk characteristics compared to the S&P 500, leading to divergent cumulative performance over time.

In 2019, the strategy showcased its strength, with a return of 25.59% , significantly outperforming the returns of all benchmarks. This trend continued in 2020, a volatile year marked by economic uncertainty, where the strategy delivered an impressive return of 131.60% , far surpassing the benchmark returns.

The proposed momentum-based clustering approach integrated with a PID controller demonstrated significant adaptability and superior performance under varying market conditions. When comparing proposed results with traditional static optimization methods, observed a CR of 135.5% over six years, which surpasses widely accepted benchmarks like S&P 500 and Russell 2000. Previous studies have employed hierarchical clustering and deep learning-based approaches to portfolio optimization, which also demonstrated improvement over conventional methods (Lu et al., 2018; Ma et al., 2020). However, these methods often lack real-time adaptability, a gap addressed by the PID controller in proposed approach. Moreover, the proposed strategy's resilience across fluctuating market environments, including periods of high volatility, is consistent with other findings on momentum-driven strategies, though our

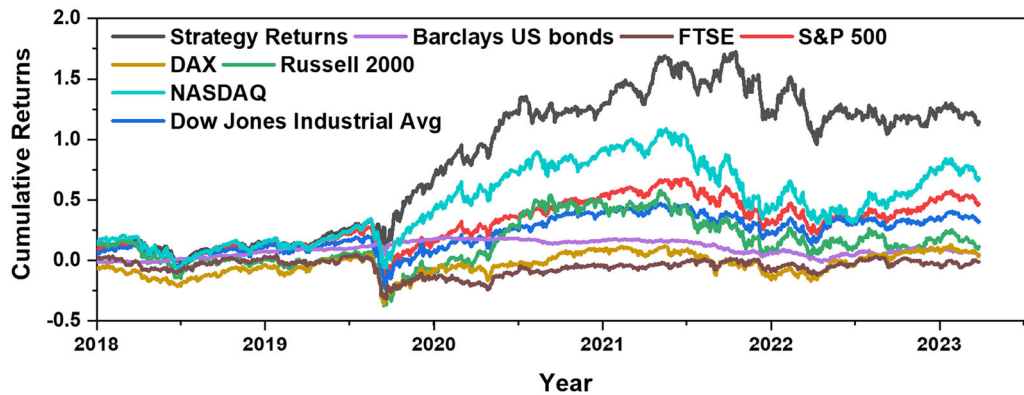


Figure 13. Compounding returns comparison against benchmarks.

Table 5. Year-wise returns comparison against benchmarks.

Year	Strategy returns	S&P 500	DJIA	Russell 2000	Barclays US bonds	DAX	NASDAQ	FTSE
2017	0.054489	0.042620	0.055133	0.027759	0.001993	-0.040208	0.031334	0.026932
2018	-0.016590	-0.047006	-0.040046	-0.135887	0.012458	-0.200896	-0.055768	-0.085302
2019	0.255895	0.215403	0.159421	0.054139	0.096562	0.011035	0.229733	0.029425
2020	1.315984	0.363350	0.263904	0.288323	0.181334	-0.034015	0.735343	-0.103380
2021	1.781319	0.669565	0.452169	0.448911	0.159964	0.077946	1.030523	-0.016045
2022	1.419235	0.311806	0.308077	0.111304	0.039037	-0.046612	0.342694	-0.038234
2023	1.355592	0.466696	0.320371	0.114746	0.050652	0.041056	0.678854	-0.012527

integrated control mechanism adds a dynamic dimension that enhances portfolio stability (Zhao et al., 2023).

Even in 2021 and 2022, as markets experienced fluctuations, the strategy remained resilient, delivering returns of 178.13% and 141.92%, respectively, consistently outpacing benchmarks. Notably, the strategy's ability to consistently deliver higher returns highlights its robustness and effectiveness in navigating various market conditions, highlighting its potential as a reliable investment option. The optimized portfolios are then evaluated using a comprehensive set of risk-adjusted performance metrics, including RB, Rolling SR, Alpha, Beta, R-Squared, MAR, Excess Return, Standard Deviation, SR, Treynor Ratio, Sortino Ratio, Jensen's Alpha and most importantly, CR. These metrics provide insights into the portfolios' performance, risk-adjusted returns, volatility and overall risk-return profile, facilitating comparisons with traditional benchmarks and validating the effectiveness of the proposed strategy.

7. Scalability, real-time adjustments and economic cycle navigation

In practical implementation, the proposed strategy faces challenges like computational demands, real-time adaptability and scalability, especially in active trading environments. To address computational demands during parameter optimization and clustering, high-performance or distributed computing solutions are recommended, as they are well-suited for the data-intensive nature of this approach. For real-time adaptability, streamlining the PID controller's tuning process is essential to enable frequent adjustments without overfitting or incurring excessive computational costs. Implementing a sliding time-window for data input can further enhance adaptability by allowing the model to dynamically respond to shifting market conditions. Scalability is another critical factor, especially when extending the approach to larger and more diverse asset pools. Scalable data processing pipelines, such as cloud-based solutions, are ideal for managing high-frequency, large-scale data. Additionally, adopting batch processing and a microservices architecture can ensure the system remains efficient and robust as asset volumes grow.

During market downturns, the proposed strategy's ability to dynamically adjust asset weights via the PID controller offers a key advantage. The controller's real-time adjustments reduce exposure to underperforming assets, effectively mitigating losses. Momentum clustering further supports this by identifying resilient stocks or those with negative momentum, enabling strategic reallocations to safer assets or defensive sectors that historically perform better during downturns.

In periods of high volatility, the strategy's adaptive nature is particularly valuable. The PID controller can swiftly respond to rapid market fluctuations, adjusting asset weights to maintain the desired risk-return profile. Momentum indicators, such as the RSI and ATR, enhance this adaptability by identifying trends and potential reversals, allowing the portfolio to capitalize on short-term price movements while effectively managing risk.

The strategy is also designed to flexibly navigate different economic cycles. During expansion phases, momentum clustering identifies growth stocks likely to outperform, while in contraction phases, the strategy shifts focus to defensive stocks or sectors. By continuously evaluating asset performance and momentum profiles, the strategy adapts to evolving economic indicators, ensuring the portfolio remains aligned with prevailing market conditions.

8. Assumptions, limitations and expanding practical applicability

In this study, the model assumes that historical data can reasonably predict future market behavior, forming the basis for using technical indicators (e.g. RSI, ATR) and the Fama-French Five Factor model for estimating RBs. However, this reliance may fail to fully capture market dynamics during structural changes or crises. The model also assumes some degree of stationarity in financial time series data, such as prices, returns and volatility. While data normalization mitigates short-term fluctuations, stationarity often breaks down during extreme events or shifting market regimes. We further assume that the K-means clustering algorithm effectively identifies homogeneous momentum clusters to inform portfolio construction. Although widely used, K-means is sensitive to the choice of cluster count (k) and may

struggle to capture meaningful groupings in markets with complex or nonlinear relationships. The PID controller adjusts portfolio weights under the assumption that proportional K_p , K_i and K_d gains can be optimally tuned for stability and responsiveness. However, optimal parameters often vary with market conditions, asset classes and time periods, necessitating frequent retuning for consistent performance.

The limitation of this model is its reliance on historical data to predict market behavior, as past performance does not always generalize well to future conditions, particularly during black swan events or economic regime shifts. While demonstrated on the S&P 500, the model may face challenges in other markets, such as emerging markets or cryptocurrencies, which exhibit higher volatility, lower liquidity or unique regulatory dynamics. Addressing these issues may require market-specific indicators or alternative asset-class models. Additionally, inaccuracies, incompleteness or noise in financial data despite interpolation techniques can introduce biases, significantly impacting performance, especially in high-frequency or small-interval data. The clustering algorithm's effectiveness depends on selecting the optimal number of clusters, while the PID controller's performance hinges on parameter tuning to balance responsiveness and stability. Suboptimal settings risk overfitting or underfitting the model. Furthermore, the assumption of minimal transaction costs and slippage may not hold in illiquid markets or during periods of high volatility, limiting real-world applicability for large portfolios or assets with constrained liquidity.

In real-world trading, factors such as transaction costs, slippage and liquidity play a significant role in determining the success of a portfolio management strategy. While proposed methodology does not explicitly incorporate these elements, we acknowledge that they are essential for practical applicability. We plan to explore how transaction costs and slippage can be integrated into our framework. For example, transaction costs could be modelled as a fixed or percentage-based fee on trades, while slippage especially in illiquid markets could be accounted for by introducing a variable cost depending on the size and timing of orders. By adjusting the portfolio's trade execution strategy, we can better simulate real-world performance and provide a more accurate assessment of the strategy's viability.

In this methodology, which integrates momentum clustering and PID control for dynamic portfolio optimization, has been applied to the S&P 500 dataset. However, we recognize the importance of extending its applicability to other asset classes and markets to enhance the generalizability of the approach. Emerging markets, which may exhibit different volatility patterns and correlations compared to developed markets, could provide an interesting test case for the proposed strategy. We plan to explore the adaptation of our methodology to these markets, adjusting the clustering and optimization parameters to reflect their unique characteristics. Furthermore, we see the potential for extending the approach to more volatile and decentralized assets such as cryptocurrencies. While the unique features of crypto markets such as high volatility and 24/7 trading pose challenges, the flexibility of our framework allows for the integration of these factors. A detailed discussion of the modifications required for each market type will be included to highlight the adaptability of our model.

9. Conclusion and future sope

This study demonstrates the transformative potential of integrating ML, momentum indicators and control system principles into portfolio management. The K-Means clustering method effectively identifies homogeneous stock groups, facilitating customized portfolio construction. By incorporating a PID controller into the optimization process, asset weights can be dynamically adjusted according to market fluctuations. The strategy's effectiveness is validated through rigorous performance analysis against major benchmarks, including the DJIA, NASDAQ and S&P 500, consistently delivering superior risk-adjusted returns. It achieved a SR of 0.577, compared to the NASDAQ's 0.29. With CR peaking at 178% in 2021 and finishing at 135%, the strategy significantly outperformed benchmarks. A Sortino Ratio of 0.837 further highlights its efficient returns relative to downside risk. This strategy through its integration of clustering techniques, momentum indicators and control systems has been empirically proven to refine portfolio management practices, securing a substantial advantage over conventional market benchmarks.

Future research could focus on enhancing adaptive PID controllers that adjust parameters in real-time to improve performance in volatile markets. Combining K-Means with alternative clustering methods, such as DBSCAN and Hierarchical Clustering, could enhance accuracy by better handling outliers and uncovering nuanced groupings. Exploring the model's application to other financial instruments including bonds, commodities, and cryptocurrencies would test its robustness across diverse markets, with tools like MPC or Fuzzy Logic Controllers offering further precision in dynamic portfolio management.

Adapting the methodology to high-frequency trading (HFT) scenarios presents another promising direction. HFT introduces challenges like large data volumes, noise, and the need for rapid decision-making. Developing faster, real-time clustering techniques and enhancing sensitivity to short-term market changes would be essential. Addressing transaction costs and liquidity measures, such as bid-ask spreads and slippage, would ensure more effective portfolio adjustments under high-frequency conditions.

Incorporating alternative data sources, such as sentiment analysis from news or social media, alongside macroeconomic indicators like interest rates and inflation, could provide a richer view of market dynamics. Additionally, integrating RL into the optimization process would allow autonomous, real-time adjustments of portfolio weights, improving adaptability and performance. By combining these innovations, future research can extend the model's applicability to various asset classes, market conditions and economic cycles, creating more robust and dynamic portfolio strategies.

For policy-making and portfolio management, dynamic control systems like PID could encourage a shift from static to adaptive strategies, potentially influencing risk management regulations. This adaptability would be especially beneficial during economic uncertainty, helping institutional investors better manage portfolios in rapidly changing conditions.

Authors' contributions

Ameiy Acharya: Conception and design of the study; analysis and interpretation of data; drafting of the manuscript; revising it critically for intellectual content; final approval of the version to be published.

Saket Gupta: Acquisition of data; analysis and interpretation of data; drafting of the manuscript; revising it critically for intellectual content; final approval of the version to be published.

Krishna Kumba: Conception and design of the study; acquisition of data; software; revising the manuscript critically for intellectual content; final approval of the version to be published.

Patri Upender: Analysis and interpretation of data; drafting of the manuscript; revising it critically for intellectual content; final approval of the version to be published.

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Human and animal rights

Humans/animals were not involved in this study.

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Data availability statement

The data used in this study was obtained from Yahoo Finance, a free and reliable financial data platform. Yahoo Finance provides open-source data accessible through its APIs, and the data collected is intended for research and educational purposes.

The nature of the data includes daily stock market information which are Open, High, Low, and Close (OHLC) prices. The data spans a period of 8 years, ending in October 2023.

The dataset comprises information from the Companies Listed in the S&P 500 Index, with four parameters (OHLC) recorded on a daily timeframe.

The data was accessed using the yfinance package, which utilizes Yahoo's publicly available APIs. This package is open-source and widely used for data retrieval and analysis in financial research.

No special rights or permissions were required to obtain this data as it is publicly available and provided by Yahoo Finance for research and educational purposes.

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